MTH644 Current Papers Questions (Fall 2021 Final Term)

① If f is measurable and $k>0$ then show that kf is measurable. ~~$\forall k \in \mathbb{R}$~~

Sol: "f" is measurable.

$\Rightarrow \{x : f(x) > a\} \in M, \quad \forall a \in \mathbb{R}$

If $k>0$, $\because f(x) > a, \quad \forall a \in \mathbb{R}$

$\Rightarrow f(x) > \frac{a}{k} \Rightarrow kf(x) > a, \quad \forall a \in \mathbb{R}$

$\therefore \{x : kf(x) > a\} \in M, \quad \forall a \in \mathbb{R}$

So "kf" is measurable.

② If f, g and $f+g$ is measurable then $f-g$ is measurable.

Sol: Given that f, g and $f+g$ is measurable.

We have to prove "$f-g$" is measurable.

$\because g$ is measurable $\Rightarrow (-1)g = -g$ is also M

$\Rightarrow f + (-1)g = f - g$

$\because f+g$ is measurable $\Rightarrow f-g$ is also measurable.

③ If $m(A)$ and $m(B)$ are infinitely measurable then prove $m(A)+m(B) = m(A \cup B) + m(A \cap B)$.

Sol: Given that $m(A) = m(B) = \infty$, then result is obvious.

$\Rightarrow m(A \cup B) = m(A \cap B) = \infty$,

So $m(A)+m(B) = m(A \cup B) + m(A \cap B)$

(proved)

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④ Find: dropped interval in contor set, for $n=1$

Sol: We know that

$$m = 2^{n-1}$$

for $n=1$, $m = 2^{1-1} = 2^0 = 1$

Hence dropped interval G_1 is

$$G_1 = I_{mm} = I_{11}$$

⑤ Define Borel Measure Space.

Sol: Restriction of lebesgue measure "m" to the class of borel set "B" is said to be Borel measure space $(\mathbb{R}, B, m)$.

⑥ $f(x) = |x+2|$, find f^+ for $f(x) \geq 0$

Sol: We know that

$$f^+ = \frac{f + |f|}{2}$$

$f = |x+2| \Rightarrow f = x+2 \quad \because f(x) \geq 0$

$|f| = ||x+2|| \Rightarrow |f| = x+2 \quad \because f(x) \geq 0$

So

$$f^+ = \frac{x+2+x+2}{2} = \frac{2x+4}{2}$$

$$f^+ = \frac{2(x+2)}{2}$$

$$f^+ = x+2 \quad \text{Ans.}$$

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$\exists$ there exists

⑦ Lebesgue measure "m" is translation invariant.

Sol: Given that say $\phi \neq E \in M$ (measurable)

$\therefore$ E is measurable $\Leftrightarrow$ for given $\varepsilon > 0$, $\exists$ an open set $U \supseteq E$, such that $m^*(U-E) < \varepsilon$

"U" is open set in $\mathbb{R} \Rightarrow$ "U+y" is also open in $\mathbb{R}$. $\forall\ y \in \mathbb{R}$

If $E \subset U \Rightarrow E+y \subseteq U+y$

$\Rightarrow (U+y)-(E+y) = (U-E)+y$

$m^*[(U+y)-(E+y)] = m^*[(U-E)+y] = m^*(U-E) < \varepsilon$

But $m^* = m \Rightarrow m[(U+y)-(E+y)] = m(U-E) < \varepsilon$

$\Rightarrow$ "E+y" is also measurable.

So "m" is translation invariant. (proved)

⑧ Let $E \subset A = [0,1)$ be a measurable set, then for each $y \in A$. The set $E \hat{+} y$ is measurable and $m(E) = m(E \hat{+} y)$.

Sol: Suppose $E_1 = E \cap [0, 1-y)$ and $E_2 = E \cap [1-y, 1)$

The set E_1, $E \cap [0, 1-y)$ and $E \cap [1-y, 1)$ are measurable

So E_1 and E_2 are also measurable. Further

$E_1 \cup E_2 = E$ and $E_1 \cap E_2 = \phi$

$\Rightarrow m(E) = m(E_1) + m(E_2) \longrightarrow (1)$ $\because$ "m" is finitely additive.

let $y_1 \in E_1$ then $y_1 < 1-y \Rightarrow y_1 + y < 1$

$\Rightarrow y_1 \hat{+} y = y_1 + y \Rightarrow E_1 \hat{+} y = E_1 + y \longrightarrow (2)$

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Now if $y_2 \in E_2 \Rightarrow \exists\, y_2 \geq 1-y \Rightarrow y_1 + y \geq 1$

$\Rightarrow y_2 \hat{+} y = y_2 + y - 1 \Rightarrow E_2 \hat{+} y = E_2 + (y-1) \longrightarrow (b)$

$\because$ m' is translation invariant.

$\Rightarrow E_1 + y$ and $E_2 + (y-1)$ are measurable.

$\Rightarrow E_1 \hat{+} y$ and $E_2 \hat{+} (y-1)$ are measurable

$\Rightarrow m(E_1 \hat{+} y) = m(E_1 + y) = m(E_1)$

and $m(E_2 \hat{+} y) = m(E_2 + (y-1)) = m(E_2)$ $\longrightarrow (2)$

$\because E_1 \cup E_2 = E$ and $E_1 \cap E_2 = \phi$

$\Rightarrow E \hat{+} y = (E_1 \hat{+} y) \cup (E_2 \hat{+} y)$ and $(E_1 \hat{+} y) \cap (E_2 \hat{+} y) = \phi$

$\Rightarrow m(E \hat{+} y) = m(E_1 \hat{+} y) + m(E_2 \hat{+} y)$ (union of two disjoint)

$= m(E_1) + m(E_2)$ by (2)

$m(E \hat{+} y) = m(E)$ by (1) (proved)

(9) A subset $\phi \neq E \subseteq R$ is measurable $\Leftrightarrow$ for given $\varepsilon > 0$ $\exists$ an open set $U \supseteq E$ with $m^*(U-E) < \varepsilon$ (for finite)

Sol: Suppose E is measurable.

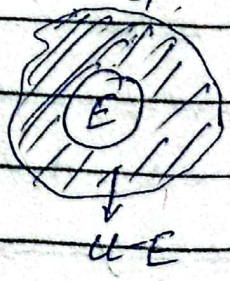
If $m^*(E) < \infty$, $\forall\, \phi \neq E \subseteq R$, for $\varepsilon > 0$, $\exists$ an open set $U \supseteq E$, $\Rightarrow m^*(U) < m^*(E) + \varepsilon \longrightarrow (a)$

$E \subseteq U \Rightarrow U = E \cup (U-E)$

$\Rightarrow m^*(U) = m^*(E) + m^*(U-E)$

$\Rightarrow m^*(U-E) = m^*(U) - m^*(E) < \varepsilon$ (from eq (a))

(proved)



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δ Delta Σ Sigma

(10) Prove that every Borel set is measurable.

Sol: Every open and closed set is measurable.
⇒ countable union or intersection of closed or open set is also measurable.
Therefore Borel set is also measurable being union or intersection of closed or open sets.

(11) Give examples of Borel set.

Sol: Examples of Borel set:

(i) $G_\delta = \bigcap_{n=1}^{\infty} u_n$ (ii) $F_\sigma = \bigcup_{n=1}^{\infty} F_n$

(12) Let $\{f_n\}$ be a sequence of extended real valued measurable function on "D" then prove that $\sup_{n\in N} (\inf_{k\geq n} f_k)$ is also measurable.

Sol: let $p = \sup_{n\in N} (\inf_{k\geq n} f_k)$ and

$G_n = \inf\{f_n, f_{n+1}, f_{n+2}, \cdots f_{n+k}, \cdots\}, \forall n \in N$

Each G_n is measurable by the measurability of f_n for each n and $\inf_{n\in N} f_n$ is measurable.
Consequently $p = \sup_{n\in N} G_n$ is a measurable function because $\sup_{n\in N} f_n$ is measurable.

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(13) Let $f: E \to R \cup \{\pm\infty\}$, $\phi \neq E \subseteq M$, where M be measurable space. Then show that:

$$\{x \in E : f(x) < a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\}$$

Sol:

Let $y \in \{x \in E : f(x) \leq a\} \to (a)$

$\Rightarrow f(y) \leq a, \forall\ a \in R.$

$\Rightarrow f(y) < a + \frac{1}{n}, \forall\ n \in R$

$\Rightarrow y \in \{x \in E : f(x) < a + \frac{1}{n}\}, \forall\ n \in R.$

$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\} \to (b)$

From eq. (a) and (b)

$$\{x \in E : f(x) \leq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) < a + \frac{1}{n}\}$$

(proved)

(14) Evaluate $m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right)$ where interval I_{nm} are pairwise disjoint and $m(I_{nm}) = l(I_{nm}) = \frac{1}{3^n}$

Sol:

Let $m(G) = m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right)$

$$= \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} m(I_{nm}) = \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} l(I_{nm})$$

$$= \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} \left(\frac{1}{3^n}\right)$$

$$= \sum_{m=1}^{2^{1-1}} \left(\frac{1}{3^1}\right) + \sum_{m=1}^{2^{2-1}} \left(\frac{1}{3^2}\right) + \sum_{m=1}^{2^{3-1}} \left(\frac{1}{3^3}\right) + \cdots$$

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$$m(G_1) = \frac{1}{3} + \left(\frac{1}{3^2} + \frac{1}{3^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3}\right) + \cdots\cdots$$

$$m(G_1) = \frac{1}{3} + \frac{2}{3^2} + \frac{4}{3^3} + \cdots\cdots$$

$$m(G_1) = 1$$

$a = \frac{1}{3}$, $r = \frac{2}{3}$

$$S_\infty = \frac{a}{1-r} = \frac{1/3}{1-2/3}$$

$$S_\infty = \frac{1/3}{1/3} = 1$$

$$m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right) = 1 \quad \text{Ans}$$

(15) Check the measurability of the cantor set in nth steps.

Sol: In general, at nth step, 2^{n-1} interval I_{nm}, $m = 1, 2, 3, \ldots 2^{n-1}$ are removed. Each I_{nm} has obviously length equal $\frac{1}{3^n} = 3^{-n}$. We have a sequence $\{I_{nm}\}$ of pairwise disjoint open sets. Thus the set $G_1 = \bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}$ being the union of a sequence of open sets is open and measurable. The compliment of G_1 with respect to $[0, 1]$ is called cantor set and is denoted by C. The set C is closed and so it must be a measurable.

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(16) If "f" and "g" are two extended real valued and measurable functions defined on same domain ($\phi \neq I$) then prove that $f \wedge g$ is a measurable.

Sol: $\because f \wedge g = \min(f, g) = \frac{1}{2}[(f+g) - |f-g|]$

So $f \wedge g$ is measurable because $(f+g)$ and $|f-g|$ are measurable and "kf" is measurable $\forall\, k \in \mathbb{R}$.

OR

$(f \wedge g)(x) = \min(f(x), g(x))$

Observe that any $a \in \mathbb{R}$, we have

$\{x : (f \wedge g)(x) > a\} = \{x : f(x) > a\} \cap \{x : g(x) > a\}$

$f(x)$ and $g(x)$ are measurable functions and the intersection of two measurable function is also measurable $\Rightarrow (f \wedge g)(x)$ is measurable.

(17) Determine the measure 1st step construction of contor set from $[0, 1]$

Sol: We know that

$$m(G_1) = m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right)$$

For 1st step

put $n = 1 \Rightarrow m = 2^{n-1} = 2^{1-1} = 2^0 = 1$

$$m(G_1) = m(I_{11}) = \ell(I_{11})$$

$$m(G_1) = m(I_{11}) = \ell(I_{11}) = \frac{1}{3^1} = \frac{1}{3} \qquad \because \ell(I_{nm}) = \frac{1}{3^n}$$

Ans

(18) Let $f: E \to \mathbb{R} \cup \{\pm\infty\}$ and $\phi \neq E \in M$, be the measurable space then prove that $\{x \in E : f(x) > 0\}$ is also measurable

Sol: By the definition of measurable function $f: E \to \mathbb{R}$ is lebesgue measurable function on $\phi \neq E \in M$ if $\{x \in E : f(x) > a\} \in M$ (measurable) for every $a \in \mathbb{R}$.

So $\{x \in E : f(x) > 0\}$ is also measurable because $0 \in \mathbb{R}$.

(19) Let $\phi \neq E \subset \mathbb{R}$, show that there exist F_σ set $F \subset E$.

Sol: For each n, let F_n be a closed set with $F_n \subseteq E$ and $m^*(E - F_n) < \frac{1}{n}$. Then $E = \bigcup_{n=1}^{\infty} F_n$ is an F_σ set $F_n \subseteq \bigcup_{n=1}^{\infty}$

$= F_n \subseteq E$ proved

(20) If $\phi = A, B \subset X$ and same member x are in $A \cup B$ Then verify that $\chi_{A \cup B} = \chi_A + \chi_B - \chi_{A \cap B}$

Sol: Given that $A, B = \phi \Rightarrow \chi_A = \chi_B = 0$

$\Rightarrow A \cup B = \phi$ and $A \cap B = \phi \Rightarrow \chi_{A \cup B} = 0$ & $\chi_{A \cap B} = 0$

Hence prove that

$\chi_{A \cup B} = 0 = \chi_A + \chi_B = \chi_{A \cap B}$

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(20) Let $\phi \neq E \subseteq \mathbb{R}$, If given that ① G_δ set "G" with $E \subseteq G$ and $m^*(G-E)=0$ then prove that ② F_σ set "F" with $F \subseteq E$ and $m^*(E-F)=0$

Sol: ① ⇒ ② ∵ $\phi \neq E \subseteq \mathbb{R} \Rightarrow \phi = E' \subseteq \mathbb{R}$

∴ For $\phi \neq E'$ given $E' \subseteq G$, $m^*(G-E')=0$

$\Rightarrow m^*(G \cap (E')') = 0$

$\Rightarrow m^*(G \cap E) = 0 \Rightarrow m^*(E \cap G) = 0$

$\Rightarrow m^*(E \cap (G')') = 0 \Rightarrow m^*(E - G') = 0$

$\Rightarrow m^*(E-F)$ where F is an F_σ set being the complement of G_δ set G. i.e. $G' = F_\sigma$

(21) Let $\phi \neq E$ be a measurable set, $f: E \to \mathbb{R} \cup \{\pm\infty\}$ then prove that $\{x \in E : f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}, \forall n \in \mathbb{N}$

Sol: Let $z \in \{x \in E : f(x) = -\infty\}$

$\Rightarrow z \in \{x \in E : f(x) < -n\}, \forall n \in \mathbb{N}$

$\Rightarrow z \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}$, "

$\Rightarrow \{x \in E : f(x) = -\infty\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\} \to (a)$

Now let $s \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}, \forall n \in \mathbb{N}$

$\Rightarrow s \in \{x \in E : f(x) < -n\}$, "

$\Rightarrow f(s) < -n \Rightarrow s \in \{x \in E : f(x) = -\infty\}$

$\Rightarrow \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\} \subseteq \{x \in E : f(x) = -\infty\} \to (b)$

From (a) and (b)

$\therefore \{x \in E : f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x \in E : f(x) < -n\}$ (proved)

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(22) If f and g are extended real valued (and measurable) functions. Then prove $\bigcup_{p\in Q}\left[\{x: f(x)>p\}\cap\{x: a-g(x)<p\}\right]\subseteq\{x: (f+g)(x)>a\}$

Sol: Let $t\in\bigcup_{p\in Q}\left[\{x: f(x)>p\}\cap\{x: a-g(x)<p\}\right]$

$\Rightarrow t\in\{x: f(x)>p\}\cap\{x: a-g(x)<p\}$ for some $p\in Q$

$\Rightarrow t\in\{x: f(x)>p\}$ and $t\in\{x: a-g(x)<p\}$

$\Rightarrow f(t)>p$ and $a-g(t)<p \Rightarrow p>a-g(t)$

$\Rightarrow f(t)>p>a-g(t) \Rightarrow f(t)>a-g(t)$

$\Rightarrow f(t)+g(t)>a \Rightarrow (f+g)(t)>a$

$\Rightarrow t\in\{x: (f+g)(x)>a\}$

So $\bigcup_{p\in Q}\left[\{x: f(x)>p\}\cap\{x: a-g(x)<p\}\right]\subseteq\{x: (f+g)(x)>a\}$ (proved)

(23) If "f" is real valued and measurable function. Then f^2 is also measurable.

Sol: We need to prove $\{x: f^2(x)>a\}\in M, \forall a\in R$

$$\{x: f^2(x)>a\}=\begin{cases}\{x: f(x)>\sqrt{a}\}\cup\{x: f(x)<-\sqrt{a}\}; & a>0\\ \text{Domain}; & a<0\end{cases}$$

$\because f$ is measurable $\Rightarrow \{x: f(x)>\sqrt{a}\}\in M$ and $\{x: f(x)<-\sqrt{a}\}\in M$

$\Rightarrow \{x: f(x)>\sqrt{a}\}\cup\{x: f(x)<-\sqrt{a}\}\in M$ and Domain is given to be measurable. i.e. Domain $\in M$

$\therefore \{x: f^2(x)>a\} \Rightarrow$ "f^2" is also measurable.

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(24) If $\phi \neq A, B \subseteq X$ Then prove that $\chi_{A-B} = \chi_A - \chi_{A\cap B}$

Sol: Case I: If $x \in A-B$

$\Rightarrow x \in A$ but $x \in B$, $x \notin A\cap B \Rightarrow \chi_{A-B} = 1$

$\chi_A - \chi_{A\cap B} = 1 - 0 = 1 = \chi_{A-B}$

Case II: If $x \notin A-B$

$$x \notin A-B = \begin{cases} x \notin A, x \notin B, x \notin A\cap B \longrightarrow (a) \\ x \notin A, x \in B, x \notin A\cap B \longrightarrow (b) \\ x \in A, x \in B, x \in A\cap B \longrightarrow (c) \end{cases}$$

$\Rightarrow \chi_{A-B} = 0$

$$\chi_A - \chi_{A\cap B} = \begin{cases} (a) \Rightarrow 0-0 = 0 \\ (b) \Rightarrow 0-0 = 0 \\ (c) \Rightarrow 1-1 = 0 \end{cases} = \chi_{A-B}$$

$\therefore$ For all cases $\chi_{A-B} = \chi_A - \chi_{A\cap B}$ (proved)

(25) Let $f: E \to \mathbb{R} \cup \{\pm\infty\}$ and $\phi \neq E \in \mathcal{M}$, where $\mathcal{M}$ be the measurable space. Then show that

$\{x \in E : f(x) \geq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$, $\forall\, n \in \mathbb{N}$

Sol: Let $y \in \{x \in E : f(x) \geq a\} \Rightarrow f(y) \geq a$, $\forall\, a \in \mathbb{R}$

$\Rightarrow f(y) > a - \frac{1}{n}$, $\forall\, n \in \mathbb{N}$

$\Rightarrow y \in \{x \in E : f(x) > a - \frac{1}{n}\}$, $\forall\, n \in \mathbb{N}$

$\Rightarrow y \in \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$

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$\Rightarrow \{x \in E : f(x) \geq a\} \subseteq \bigcap_{n=1}^{\infty} \{x \in E : f(x) > a - \frac{1}{n}\}$ (proved)

(26) For a given $\phi \neq E \subset R$ and $\varepsilon > 0$, there is a a closed set $F \subset E$ such that $m^*(E-F) < \varepsilon$, then show that $\exists$ an open set satisfying $m^*(O-E) < \varepsilon$.

Sol: For $\phi \neq E \subset R \Rightarrow E' \subset R$ .. $\exists$ a closed set "F" and $\varepsilon > 0$ such that $F \subset E'$ and $m^*(E'-F) < \varepsilon \rightarrow (1)$

$F \subset E' \Rightarrow E = E'' \subset E' = O$ (open set) say

$\Rightarrow m^*(O-E) = m^*(O \cap E') = m^*(E' \cap O)$

$\Rightarrow m^*(O-E) = m^*(E'-O') = m^*(E'-F) < \varepsilon$ from (1)

$\Rightarrow m^*(O-F) < \varepsilon$ (proved)

(27) If $m^*(G-E) = 0$, then show that $m^*(G-E^c) = 0$ (2)

Sol: $\because m^*(G-E) = 0$, $\phi \neq E \subset R \Rightarrow \phi \neq E' \subseteq R$

$\therefore$ for $\phi \neq E'$ gives $E' \subseteq G$ $\quad \therefore E \subseteq G$

Hence $m^*(G-E^c) = 0$ (proved)

(28) Let f be an extended real valued measurable function on D and $\phi \neq A \subset D$. Then show that $\{x \in A : f(x) > a\}$ is measurable. $\forall a \in R$ (3)

For Solution See Question No. 36

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(29) Let "f" be an extended real valued and measurable functions defined on some domain $\phi \neq D$ then prove that $|f|$ is measurable function.

Sol: ∵ "f" is measurable $\Rightarrow \{x : f(x) > a\} \in M, \forall a \in \mathbb{R}$

To prove $|f|$ is measurable, we need to show that

$$\{x : |f|(x) > a\} \in M, \forall a \in \mathbb{R}$$

$$\{x : |f|(x) > a\} = \{x : |f(x)| > a\}$$

$$\{x : |f|(x) > a\} = \begin{cases} \{x : f(x) > a\} \cup \{x : f(x) < -a\} ; & a > 0 \\ D ; & a \leq 0 \end{cases}$$

In all cases on R.H.S, we get a measurable sets.

$\Rightarrow \{x : |f|(x) > a\}$ is measurable $\forall a \in \mathbb{R}$

Therefore $|f|$ is also measurable.

(30) Countable set are Borel sets with zero measure.

Sol: Case-I Singleton

$$\forall a \in \mathbb{R} ; \{a\} = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, a + \frac{1}{n}\right) ; n \in \mathbb{N}$$

∵ $\left\{\left\{a - \frac{1}{n}, a + \frac{1}{n}\right\}, \downarrow\right\}$ is decreasing sequence.

with $m(E_n) = \frac{2}{n} < \infty$ ∴ where $E_n = \left(a - \frac{1}{n}, a + \frac{1}{n}\right)$

$$\Rightarrow m\{a\} = m\left\{\bigcap_{n=1}^{\infty} E_n\right\} = \lim_{n \to \infty} m(E_n) = \lim_{n \to \infty} m(E_n)$$

$$\Rightarrow m\{a\} = \lim_{n \to \infty} \left(\frac{2}{n}\right) = 0$$

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Case II : Finite set

let $\{x_1, x_2, x_3, \ldots x_n\} = \bigcup_{i=1}^{n} \{x_i\}$

$\Rightarrow m\left(\bigcup_{i=1}^{n} \{x_i\}\right) = \sum_{i=1}^{n} m\{x_i\} = 0$

Case III : Infinite set

Let $\{x_1, x_2, x_3, \ldots\} = \bigcup_{i=1}^{\infty} \{x_i\}$

$\Rightarrow m\left(\bigcup_{i=1}^{\infty} \{x_i\}\right) = \sum_{i=1}^{\infty} m\{x_i\} = 0$

(proved)

(31) Open interval $(a,b) \subset \mathbb{R}$ is measurable.

Sol: (a,∞) is measurable.

$\Rightarrow (a,\infty)^c = (-\infty, a]$ is also measurable.

and $(-\infty, b-\frac{1}{n}]$ is also measurable $\forall n \in \mathbb{N}$, as $n \to \infty$

and $(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b-\frac{1}{n}]$ is also measurable.

because being countable union of measurable set

$\Rightarrow (a,\infty) \cap (-\infty, b) = (a,b)$ is also measurable.

because being intersection of two measurable set.

(32) If $A = \bigcup_{n=1}^{\infty} A_n$ and $\{A_n\}_{n=1}^{\infty}$ is a sequence of pairwise disjoint subsets of X. Then shows that $\chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$

Sol: Case I: If $x \in A \Rightarrow x \in \bigcup_{n=1}^{\infty} A_n$

$\Rightarrow x \in A_k$ exactly for $n=k$

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$x \notin \bigcup_{n=1}^{\infty} A_n = \{A_k\}$

$\Rightarrow \chi_{A_k}(x) = 1 \;,\; \chi_{A_m}(x) = 0, \;\forall\; n \neq k$

$x \in A \Rightarrow \chi_A(x) = 1$

$\therefore \chi_A(x) = 1 = 0+0+\cdots+\underset{n=k}{1}+0+0+\cdots$

$\chi_A(x) = 1 = \sum_{n=1}^{\infty} \chi_{A_n}(x) \Rightarrow \chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$

Case II: $x \notin A = \bigcup_{n=1}^{\infty} A_n \Rightarrow x \notin A_n, \forall n \in \mathbb{N}$

$\Rightarrow \chi_A(x) = 0 \quad | \quad \Rightarrow \chi_{A_n}(x) = 0, \forall n \in \mathbb{N}$

$\chi_A(x) = 0+0+0+\cdots+0+\cdots = \sum_{n=1}^{\infty} \chi_{A_n}(x)$

$\Rightarrow \chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$ (proved)

Ex: (33) Let $\{E_n, \uparrow\}_{n=1}^{\infty}$ be an increasing sequence of measurable sets. Then show that

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) = m\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_n)$$

Sol: Given that $(E_n, \uparrow)$, such that

$\Rightarrow E_1 \subset E_2 \subset E_3 \subset \cdots$ of measurable sets.

Defining: $F_1 = E_1, F_2 = E_2 - E_1, F_3 = E_3 - F_2, \ldots$

$F_n = E_n - E_{n-1}, \ldots$ where $\{F_n\}_{n=1}^{\infty}$ is a sequence of disjoint measurable sets

$\Rightarrow \bigcup_{k=1}^{n} F_k = \bigcup_{k=1}^{n} E_k \Rightarrow \bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k$

$\therefore m^*\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} F_k\right) = m\left(\bigcup_{k=1}^{\infty} E_k\right)$

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$$m^*\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} m(E_k) = \lim_{n\to\infty} \sum_{k=1}^{\infty} m(E_k)$$

$$= \lim_{n\to\infty} m\left(\bigcup_{k=1}^{\infty} E_k\right) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{\infty} E_k\right)$$

$$= \lim_{n\to\infty} m^*(E_n) \qquad \text{(proved)}$$

(34)

Let $\{E_n, \downarrow\}_{n=1}^{\infty}$ be a decreasing sequence of measurable sets such that $m(E_k) < \infty$ for some $k \in \mathbb{N}$. Then shows that $m^*\left(\bigcap_{n=1}^{\infty} E_n\right) = m\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_n)$

Sol: Let $m(E_1) < \infty$, say

$\because \langle E_n \downarrow \rangle \Rightarrow E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots\cdots$

$\Rightarrow \phi = E_1 - E_1 \subseteq E_1 - E_2 \subseteq E_1 - E_3 \subseteq \cdots\cdots E_1 - E_n \subseteq \cdots\cdots$

$\Rightarrow m^*\left(\bigcup_{n=1}^{\infty} (E_1 - E_n)\right) = \lim_{n\to\infty}{}^{m^*}(E_1 - E_n)$

$$\bigcup_{n=1}^{\infty} (E_1 - E_n) = \bigcup_{n=1}^{\infty} (E_1 \cap E_n')$$

$$= (E_1 \cap E_1') \cup (E_2 \cap E_2') \cup \cdots\cdots$$

$$= E_1 \cap [E_1' \cup E_2' \cup E_3' \cup \cdots]$$

$$= E_1 \cap [E_1 \cap E_2 \cap E_3 \cap \cdots]'$$

$$= E_1 \cap \left(\bigcap_{n=1}^{\infty} E_n\right)'$$

$$\bigcup_{n=1}^{\infty} (E_1 - E_n) = E_1 - \left(\bigcap_{n=1}^{\infty} E_n\right)$$

$$\lim_{n\to\infty} m^*(E_1 - E_n) = m^*\left(E_1 - \bigcap_{n=1}^{\infty} E_n\right)$$

$$\lim_{n\to\infty} m^*(E_1) - \lim_{n\to\infty}{}^{m^*}(E_n) = m^*(E_1) - m^*\left(\bigcap_{n=1}^{\infty} E_n\right)$$

$$m^*(E_1) - \lim_{n\to\infty}{}^{m^*}(E_n) = m^*(E_1) - m^*\left(\bigcap_{n=1}^{\infty} E_n\right)$$

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$\Rightarrow m^*\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m^*(E_n)$ (proved)

(3a) If "f" and "g" are two extended real valued and measurable functions defined on same domain ($\phi \neq D$) then prove that $f \vee g = \max(f, g)$ is measurable.

Sol: $\because f \vee g = \max(f, g) = \frac{1}{2}[(f+g) + |f-g|]$

So $f \vee g$ is measurable because $(f+g)$ and $|f-g|$ are measurable and kf is measurable for all $k \in R$.

OR

$(f \vee g)(x) = \max(f(x), g(x))$

For any $a \in R$, we have

$\{x : (f \vee g)(x) > a\} = \{x : f(x) > a\} \cup \{x : g(x) > a\}$

$f(x)$ and $g(x)$ are measurable functions and the Union of two measurable functions is also measurable.

$\Rightarrow (f \vee g)(x)$ is measurable.

(3b) Let f be an extended real valued measurable function on D and $\phi \neq A \subset D$. Then show that restriction of "f" to A is also measurable and $A \in M$. (OR). $\{x \in A : f_A(x) > a\}$ is measurable. $\forall a \in R$.

Sol: $\because$ "f" is measurable

$\rightarrow \{x \in D : f(x) > a\} \in M$, $\forall a \in R$.

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To prove "f_A" is measurable we have
$\{x \in A : f_A(x) > a\} \in \mathcal{M}$.

Note: $\{x \in A : f_A(x) > a\} = \{x \in D : f(x) > a\} \cap A$

On R.H.S both sets are measurable.

$\therefore$ Intersection of two measurable sets is measurable

$\Rightarrow$ "f_A" is measurable; $\{x \in A : f_A(x) > a\} \in \mathcal{M}$

(proved)

(37) Show that $f = f^+ - f^-$

Sol: If $f(x) \geq 0 \Rightarrow f^+(x) = f(x)$, $f^-(x) = 0$

$\therefore (f^+ - f^-)(x) = f^+(x) - f^-(x) = f(x) - 0 = f$

If $f(x) < 0 \Rightarrow f^+(x) = 0$, $f^-(x) = -f(x)$

$\therefore (f^+ - f^-)(x) = f^+(x) - f^-(x) = 0 - (-f(x)) = f$

$\Rightarrow f^+ - f^- = f$ (proved)

(38) Show that $f^+ = \frac{f + |f|}{2}$

Sol: If $f(x) \geq 0 \Rightarrow f^+(x) = f(x) \Rightarrow f^+ = f$

$|f(x)| = |f|(x) = f(x)$

$\therefore \frac{f + |f|}{2}(x) = \frac{f(x) + |f|(x)}{2} = \frac{f(x) + f(x)}{2} = f(x) = f^+(x)$

If $f(x) < 0 \Rightarrow f^+(x) = 0$, $\therefore |f|(x) = |f(x)| = -f(x)$

$\therefore \frac{f + |f|}{2}(x) = \frac{f(x) + |f|(x)}{2} = \frac{f(x) - f(x)}{2} = 0 = f^+(x)$

$\Rightarrow \frac{f + |f|}{2} = f^+$ (proved)

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Solved by Mahar Afaq Safdar Muhammadi

دعاؤں میں یاد رکھیے گا۔ جزاک اللہ

(39) Define Borel set.

Sol: A set "E" is said to be a Borel set if it can be expressed as countable union or intersection of open or closed sets.

e.g. $\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x + \frac{1}{n}\right)$ is Borel set.

↳ G_δ

(40) Define Sum Modulo 1 and Translation Modulo 1.

Sol: Sum Modulo 1:

Let $x, y \in A = [0, 1)$ and we define sum modulo 1 of x and y as

$$x \hat{+} y = \begin{cases} x + y & \text{if } x + y < 1 \\ x + y - 1 & \text{if } x + y \geq 1 \end{cases}$$

Translation Modulo 1.

Let $E \subseteq A = [0, 1)$ and $y \in A$, we define a translation modulo 1 of "E" as

$$E \hat{+} y = \{x \hat{+} y, x \in E\}$$

$\overline{E}$:

Set of all Sum Modulo 1.

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(41) Show that $|f| = f^+ + f^-$.

Sol. If $f(x) \geq 0 \Rightarrow f^+(x) = f(x)$, $f^-(x) = 0$

$\therefore (f^+ + f^-)(x) = f^+(x) + f^-(x) = f(x) + 0 = f(x) = |f(x)|$

If $f(x) < 0 \Rightarrow f^+(x) = 0$, $f^-(x) = f(x)$

$\therefore (f^+ + f^-)(x) = f^+(x) + f^-(x) = 0 + f(x) = f(x) = |f(x)|$

$\Rightarrow |f| = f^+ + f^-$ (proved)

(42) Show that $\min(f,g) = \dfrac{(f+g) - |f-g|}{2}$

Sol:

$$|f-g|(x) = |f(x) - g(x)| = \begin{cases} f(x) - g(x) & ; f(x) \geq g(x) \\ g(x) - f(x) & ; f(x) < g(x) \end{cases}$$

Case I:

If $f(x) \geq g(x) \Rightarrow f(x) - g(x) \geq 0$

$\Rightarrow |f(x) - g(x)| = f(x) - g(x)$ and

If $f(x) \geq g(x) \Rightarrow \min(f,g) = g(x)$

$$\therefore \frac{(f+g) - |f-g|}{2}(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}$$

$$= \frac{f(x) + g(x) - (f(x) - g(x))}{2}$$

$$= \frac{f(x) + g(x) - f(x) + g(x)}{2} = \frac{2g(x)}{2} = g(x)$$

$$\Rightarrow \frac{(f+g) - |f-g|}{2} = \min(f,g)$$

Case II: If $f(x) < g(x) \Rightarrow f(x) - g(x) < 0$

$\Rightarrow |f(x) - g(x)| = g(x) - f(x)$ and

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If $f(x) < g(x) \Rightarrow \min(f,g) = f(x)$

$\therefore \frac{(f+g)-|f-g|}{2}(x) = \frac{f(x)+g(x)-|f(x)-g(x)|}{2}$

$= \frac{f(x)+g(x)-(g(x)-f(x))}{2}$

$= \frac{f(x)+g(x)-g(x)+f(x)}{2} = \frac{2f(x)}{2} = f(x)$

$\Rightarrow \frac{(f+g)-|f-g|}{2} = \min(f,g)$ (proved)

(43) Show that $\max(f,g) = \frac{(f+g)-|f-g|}{2}$

Sol:

$$|f-g|(x) = |f(x)-g(x)| = \begin{cases} f(x)-g(x); & f(x) \geq g(x) \\ g(x)-f(x); & f(x) < g(x) \end{cases}$$

Case I: If $f(x) \geq g(x) \Rightarrow f(x)-g(x) \geq 0$

$\Rightarrow |f(x)-g(x)| = f(x)-g(x)$ and

If $f(x) \geq g(x) \Rightarrow \max(f,g) = f(x)$

$\therefore \frac{(f+g)+|f-g|}{2}(x) = \frac{f(x)+g(x)+|f(x)-g(x)|}{2}$

$= \frac{f(x)+g(x)+f(x)-g(x)}{2} = \frac{2f(x)}{2} = f(x)$

$\Rightarrow \frac{(f+g)+|f-g|}{2} = \max(f,g)$

Case II: If $f(x) < g(x) \Rightarrow f(x)-g(x) < 0$

$\Rightarrow |f(x)-g(x)| = g(x)-f(x)$ and

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If $f(x) \le g(x) \Rightarrow \max(f,g) = g(x)$

$\therefore \dfrac{(f+g)+|f-g|}{2}(x) = \dfrac{f(x)+g(x)+|f(x)-g(x)|}{2}$

$= \dfrac{f(x)+g(x)+g(x)-f(x)}{2} = \dfrac{2g(x)}{2} = g(x)$

$\Rightarrow \dfrac{(f+g)+|f-g|}{2} = \max(f,g)$ (proved)

(44) Show that $\chi_{A\cap B} = \chi_A \cdot \chi_B$

Sol: If $x \in A\cap B \Rightarrow x \in A$ and $x \in B$

$\Rightarrow \chi_{A\cap B} = 1 = \chi_A = \chi_B = 1.1 = \chi_A \cdot \chi_B$

If $x \notin A\cap B \Rightarrow$
- $x \in A$ but $x \notin B \Rightarrow \chi_A = 1, \chi_B = 0$
- $x \in B$ but $x \notin A \Rightarrow \chi_B = 1, \chi_A = 0$
- $x \notin A$ and $x \notin B \Rightarrow \chi_A = 0, \chi_B = 0$

$x \notin A\cap B \Rightarrow \chi_{A\cap B} = 0 \quad \{\because \chi_A \cdot \chi_B = 0$

For all, $x \notin A\cap B$; $\chi_{A\cap B} = 0 = \chi_A \cdot \chi_B$

$\Rightarrow \chi_{A\cap B} = \chi_A \cdot \chi_B$ (proved)

(45) Define open set in $\mathbb{R}$.

Sol: Any open set in $\mathbb{R}$ say "u" can be written as countable union of disjoint open intervals i.e. $u = \bigcup_{n=1}^{\infty} \{I_n\}$ and each open intervals is measurable and countable union of measurable set is measurable.

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(46) If $f(x) = \cos x$, Find f^+, $-\pi \le x \le \pi$

Sol: We know that

$$f^+(x) = \frac{f + |f|}{2}(x) = \frac{f(x) + |f|(x)}{2}$$

$\because f(x) = \cos x$

$$|f|(x) = |f(x)| = \begin{cases} \cos(x) & ; \ -\frac{\pi}{2} \le x \le \frac{\pi}{2} \\ -\cos(x) & ; \ \frac{\pi}{2} < x \le \pi, \ -\pi \le x < \frac{\pi}{2} \end{cases}$$

$$f^+(x) = \begin{cases} \frac{\cos x + \cos x}{2} = \cos x & ; \ -\frac{\pi}{2} \le x \le \frac{\pi}{2} \\ \frac{\cos x - \cos x}{2} = 0 & ; \ \frac{\pi}{2} < x \le \pi, \ -\pi \le x < -\frac{\pi}{2} \end{cases}$$

(47) Let $\{E_i\}_{i=1}^n$ be a sequence in a measurable space $\mathcal{M}$, the m is countably additive.

Sol: $\forall \ \phi \neq A \subseteq \mathbb{R}$ and for the sequence $\{E_i\}_{i=1}^n$ of pairwise disjoint measurable sets:

$$m^*\left(A \cap \left(\bigcup_{i=1}^{\infty} E_i\right)\right) = \sum_{i=1}^{\infty} m^*(A \cap E_i)$$

put $A = \mathbb{R}$ $\Rightarrow$ $m^*\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m^*(E_i)$

$$\Rightarrow m\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m(E_i)$$

Hence m is countably additive.

Note: If the question asked for "finitly additive" then in this solution replaced "∞" by "n".

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(48) Let $\{E_i\}_{i=1}^{n}$ be a sequence in a measurable space $\mathcal{M}$, then m is monotonic.

Sol: Let $\phi \neq A, B \in \mathcal{M}$, such that $B \subseteq A$

$\Rightarrow A = B \cup (A-B)$

$\because m^*$ is finitely additive.

$\Rightarrow m^*(A) = m^*(B) + m^*(A-B)$

$\Rightarrow m^*(A) \geq m^*(B) \quad \because m^*(A-B) \geq 0$

$\Rightarrow m^*(B) \leq m^*(A)$

(Proved)

(49) Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of extended real valued measurable function on D. $f = \inf_{n \in \mathbb{N}} \{f_n\}$, further if $\{x : f(x) > a\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\}$, $\forall a \in \mathbb{R}$, then show that f is measurable. (2)

Sol: Given that $f = \inf_{n \in \mathbb{N}} \{f_n\}$ and

$\{x : f(x) > a\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) > a\}$; $\forall a \in \mathbb{R}$

The intersection of a countable number of measurable sets is measurable. So f must be measurable.

(50) In the construction of cantor set $[0,1]$, evaluate the measure of dropped interval in the first step (case $n=1$) (2)

Sol: We know that $m = 2^{n-1} = 2^{1-1} = 2^0 = 1$, $\because n=1$

$m(G_1) = m(I_{mm}) = m(I_{11}) = l(I_{11}) = \frac{1}{3}$

$\because m[0,1] = G \cup C \Rightarrow m[0,1] = m(G_1) + m(C)$

$\Rightarrow m(C) = m[0,1] - m(G_1) = (1-0) - \frac{1}{3} = 1 - \frac{1}{3} = \frac{2}{3}$ Ans

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(51) Show that $f^- = \frac{|f| - f}{2}$

Sol: If $f(x) \geq 0 \Rightarrow f^-(x) = 0$, $|f(x)| = |f|(x) = f(x)$

$\therefore \frac{|f| - f}{2}(x) = \frac{|f|(x) - f(x)}{2} = \frac{f(x) - f(x)}{2} = 0 = f^-(x)$

If $f(x) < 0 \Rightarrow f^-(x) = -f(x)$, $|f(x)| = |f|(x) = -f(x)$

$\therefore \frac{|f| - f}{2}(x) = \frac{|f|(x) - f(x)}{2} = \frac{-f(x) - f(x)}{2} = -f(x) = f^-(x)$

$\Rightarrow \frac{|f| - f}{2} = f^-$ (proved)

(52) If $f(x) = \sin x$ Find f^+ and f^-, $-\pi \leq x \leq \pi$

Sol: We know that

$$f^+(x) = \frac{f + |f|}{2}(x) = \frac{f(x) + |f|(x)}{2}$$

$\because f(x) = \sin x$

$$|f|(x) = |f(x)| = \begin{cases} \sin x & ; \ 0 \leq x \leq \pi \\ -\sin x & -\pi < x \leq -\frac{\pi}{2}, \ -\frac{\pi}{2} < x < 0 \end{cases}$$

$$f^+(x) = \frac{f(x) + |f|(x)}{2} = \begin{cases} \frac{\sin x + \sin x}{2} = \sin x & ; \ 0 \leq x \leq \pi \\ \frac{\sin x - \sin x}{2} = 0 & ; \ -\pi < x \leq -\frac{\pi}{2}, \ -\frac{\pi}{2} < x < 0 \end{cases}$$

We know that $f^-(x) = \frac{|f| - f}{2}(x) = \frac{|f|(x) - f(x)}{2}$

$$f^-(x) = \frac{|f|(x) - f(x)}{2} = \begin{cases} \frac{\sin x - \sin x}{2} = 0 ; & 0 \leq x \leq \pi \\ \frac{-\sin x - \sin x}{2} = -\sin x ; & -\pi < x \leq \frac{-\pi}{2}, \ -\frac{\pi}{2} < x < 0 \end{cases}$$

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(53) If $f(x)=\cos x$, Find f^-, $-\pi \le x \le \pi$

Sol: We know that

$$f^-(x)=\frac{|f|-f(x)}{2}=\frac{|f|(x)-f(x)}{2}$$

$\because f(x)=\cos x$

$$|f|(x)=|f(x)|=\begin{cases}\cos x & ; \frac{-\pi}{2}\le x\le\frac{\pi}{2}\\ -\cos x & ; \frac{\pi}{2}\le x\le\pi,\ -\pi\le x<\frac{-\pi}{2}\end{cases}$$

$$f^-(x)=\frac{|f|(x)-f(x)}{2}=\begin{cases}\frac{\cos x-\cos x}{2}=0 & ; \frac{-\pi}{2}\le x\le\frac{\pi}{2}\\ \frac{-\cos x-\cos x}{2}=-\cos x & ; \frac{\pi}{2}<x\le\pi,\ -\pi\le x<\frac{-\pi}{2}\end{cases}$$

(54) Evaluate $E\hat{+}\frac{1}{5}$, the translation modulo 1 of the set $E=\left\{0,\frac{1}{2},\frac{1}{3},\frac{2}{3}\right\}\subset[0,1[$ by $\frac{1}{5}$.

Sol. We know that $E\hat{+}y=\{x\hat{+}y\ ,\ x\in E\}$

$y=\frac{1}{5}$

$$E\hat{+}\frac{1}{5}=\left\{0\hat{+}\frac{1}{5},\frac{1}{2}\hat{+}\frac{1}{5},\frac{1}{3}\hat{+}\frac{1}{5},\frac{2}{3}\hat{+}\frac{1}{5}\right\}$$

$$E\hat{+}\frac{1}{5}=\left\{0+\frac{1}{5},\frac{1}{2}+\frac{1}{5},\frac{1}{3}+\frac{1}{5},\frac{2}{3}+\frac{1}{5}\right\}$$

$\because x\hat{+}y=\begin{cases}x+y\ ; x+y<1\\ x+y-1\ ; x+y\ge1\end{cases}$

$$E\hat{+}\frac{1}{5}=\left\{\frac{1}{5},\frac{7}{10},\frac{8}{15},\frac{13}{15}\right\}$$ Ans

(55) In $[0,1[$, the sum modulo 1, $x\hat{+}y$ of $\frac{5}{2}$ and $\frac{1}{2}$ = ... (MCQs)

Sol: $x\hat{+}y=\begin{cases}x+y\ ; x+y<1\\ x+y-1\ ; x+y\ge1\end{cases}$ ✓

$\begin{cases}\frac{5}{2}+\frac{1}{2}=\frac{6}{2}=3>1\end{cases}$

$x\hat{+}y=\frac{5}{2}+\frac{1}{2}-1=3-1=2$ Ans

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(56) An extended real valued function defined on a set of measure zero is measurable.

Sol: Let $\phi \neq E \subseteq \mathbb{R}$, such that $m^*(E) = 0$ and $f: E \to \mathbb{R} \cup \{+\infty\}$, To prove f is measurable, we need to show that $\{x \in E : f(x) > a\} \in M$, $\forall\, a \in \mathbb{R}$.

$\because \{x \in E : f(x) > a\} \subseteq E$

$\Rightarrow m^*\{x \in E : f(x) > a\} \leq m^*(E)$ $\because$ By monotocity of "m^*"

$\Rightarrow m^*\{x \in E : f(x) > a\} \leq 0 \rightarrow$ (a) (given)

But by non-negativity of "m^*"

$\Rightarrow m^*\{x \in E : f(x) > a\} \geq 0 \rightarrow$ (b)

From (a) and (b) $\Rightarrow m^*\{x \in E : f(x) > a\} = 0$, $\forall\, a \in \mathbb{R}$.

$\Rightarrow \{x \in E : f(x) > a\} \in M$, $\forall\, a \in \mathbb{R}$.

So f is measurable. (proved)

(57) If "f" is measurable and $k < 0$, then show that kf is measurable.

Sol: $\because f$ is measurable.

$\Rightarrow \{x : f(x) > a\} \in M$, $\forall\, a \in \mathbb{R}$

If $k < 0$, $f(x) > a$, $\forall\, a \in \mathbb{R}$ (given)

$\Rightarrow f(x) < \frac{a}{k} \Rightarrow kf(x) < a$ $\because (k < 0)$

$\Rightarrow \{x : kf(x) < a\} \in M$, $\forall\, a \in \mathbb{R}$

$\Rightarrow$ "kf" is measurable.

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(58) Let $f: E \to \mathbb{R} \cup \{\pm\infty\}$, $\phi \neq E \in M$, If $\{x \in E : f(x) \geq a\}$ is measurable, then $\{x \in E : f(x) \geq a\}$ is measurable for all $a \in \mathbb{R}$.

Sol: Given that $\{x \in E : f(x) \geq a\} \in M$, $\forall a \in \mathbb{R}$.

We know that

$$\{x \in E : f(x) < a\} = \{x \in E : f(x) \geq a\}^c, \forall a \in \mathbb{R}$$

$\because \{x \in E : f(x) \geq a\} \in M \Rightarrow \{x \in E : f(x) \geq a\}^c \in M$

Hence $\Rightarrow \{x \in E : f(x) < a\} \in M$, $\forall a \in \mathbb{R}$

(proved)

(59) Let $f: E \to \mathbb{R} \cup \{\pm\infty\}$, $\phi \neq E \in M$, If $\{x \in E : f(x) \leq a\}$ is measurable, the $\{x \in E : f(x) > a\}$ is measurable for all $a \in \mathbb{R}$.

Sol:

Given that $\{x \in E : f(x) \leq a\} \in M$, $\forall a \in \mathbb{R}$.

We know that

$$\{x \in E : f(x) > a\} = \{x \in E : f(x) \leq a\}^c, \forall a \in \mathbb{R}.$$

$\because \{x \in E : f(x) \leq a\} \in M \Rightarrow \{x \in E : f(x) \leq a\}^c \in M$.

$\Rightarrow \{x \in E : f(x) > a\}$ is measurable, $\forall a \in \mathbb{R}$.

(proved)

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(60) Check the measurability of ~~that~~ cantor set in nth steps.

Sol: "G" is open being countable union of disjoint open intervals $\Rightarrow$ "G" is measurable.

$\because C = G' = [0,1] - G$ closed and measurable.

$\therefore G = \bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}$

$$m(G) = m\left(\bigcup_{n=1}^{\infty} \bigcup_{m=1}^{2^{n-1}} I_{nm}\right)$$

$\because$ m is countably additive.

$$m(G) = \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} m(I_{nm})$$

$$m(G) = \sum_{n=1}^{\infty} \sum_{m=1}^{2^{n-1}} \ell(I_{nm})$$

expanding firstly by "n"

$$m(G) = \sum_{m=1}^{2^{1-1}} \ell(I_{1m}) + \sum_{m=1}^{2^{2-1}} \ell(I_{2m}) + \sum_{m=1}^{2^{3-1}} \ell(I_{3m}) + \cdots$$

$$m(G) = \underbrace{\ell(I_{11})}_{} + \underbrace{\ell(I_{21}) + \ell(I_{22})}_{} + \underbrace{\ell(I_{31}) + \ell(I_{32}) + \ell(I_{33}) + \ell(I_{34})}_{} + \cdots$$

$$m(G) = \frac{1}{3} + \left(\frac{1}{3^2} + \frac{1}{3^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3}\right) + \cdots \quad \because \ell(I_{nm}) = \frac{1}{3^n}$$

$$m(G) = \frac{1}{3} + \frac{2}{3^2} + \frac{4}{3^3} + \cdots$$

$$a = \frac{1}{3}, \quad r = \frac{2}{3}$$

$$S_\infty = \frac{a}{1-r} = \frac{1/3}{1-2/3} = \frac{\frac{1}{3}}{\frac{1}{3}} = 1$$

$$\boxed{m(G) = 1}$$

$\because [0,1] = G \cup C$

$\Rightarrow m[0,1] = m(G) + m(C)$

$\Rightarrow 1 = 1 + m(C)$

$\Rightarrow \boxed{m(C) = 0}$

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